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Project TITLE

RF Field Strength Meter

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LIST OF ABBREVIATIONS

VHF	Very High Frequency
UHF	Ultra High Frequency
FSM	Field Strength Meter
RF	Radio Frequency
LED	Light-Emitting Diode

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INTRODUCTION

1.1. Abstract

The purpose of this project was to design and implement a portable, wideband Radio Frequency (RF) Field Strength Meter (FSM) capable of detecting electromagnetic radiation in the VHF and UHF bands. The device utilizes a voltage-doubler detection stage consisting of 1N60 Germanium diodes, chosen specifically for their low forward voltage drop of approximately 0.3V which allows for the rectification of weak RF signals that standard silicon diodes would fail to capture. To provide a readable output, a two-stage amplification system was integrated, utilizing a 2N4401 NPN transistor followed by a BC516 PNP Darlington pair to achieve high current gain. The output is displayed via a four-level LED indicator network, where each LED corresponds to a specific signal intensity threshold: Low, Medium, High, and Full⁴. Experimental results confirmed that the meter successfully responds to frequencies such as 146 MHz (VHF) and 443 MHz (UHF), with LED activation patterns correlating predictably to the proximity of the RF source. This project demonstrates a cost-effective solution for RF troubleshooting, antenna alignment, and interference detection.

1.2. Project overview

A Field Strength Meter (FSM) is a fundamental instrument in radio engineering, used to detect and quantify the intensity of electromagnetic radiation (RF signals) present in the surrounding environment. Such meters are widely employed in antenna alignment, transmitter testing, interference detection, and general RF diagnostics.

This project presents the design, construction, and evaluation of a simple, low-cost, wideband RF Field Strength Meter based primarily on bipolar junction transistors (BJTs) and a Germanium diode detector. The proposed meter provides a visual representation of received signal strength using a four-level LED

indicator corresponding to Low, Medium, High, and Full RF levels. The device operates from a 12 V DC supply and is suitable for detecting VHF and UHF signals.

The motivation behind this project is to develop an accessible and practical RF troubleshooting tool that allows users to:

- Understand basic RF detection principles
- Gain hands-on experience with transistor-based amplification stages

1.3. Theoretical Introduction

1.3.1. Principles of RF Detection

The operation of the FSM relies on two key principles: RF rectification and DC amplification.

1.3.1.1. Rectification (Detection)

RF energy captured by the antenna appears as a high-frequency AC voltage. This AC signal must be converted into a proportional DC value before it can be interpreted. Two 1N60 Germanium diodes are used for detection due to their low forward voltage drop (~ 0.3 V), allowing the circuit to detect weak RF signals. Silicon diodes (~ 0.7 V drop) would fail to rectify such low-level inputs.

1.3.1.2. Amplification

The rectified DC voltage is very small, typically too small to drive LEDs directly. To address this, a two-stage amplifier is implemented:

1. Stage 1 (2N4401 NPN): Provides initial current amplification from the detector.
2. Stage 2 (BC516 PNP Darlington): Offers very high current gain (β), allowing small input variations to produce noticeable LED brightness differences.

This combination yields a sensitive, low-noise amplification system capable of responding to weak RF fields.

Table 1:Component List and Functions.

Component	Value / Type	Function
Antenna	6 inches (initial)	RF signal capture
Detector Diodes	2 × 1N60	RF rectification
Transistor Q1	2N4401 (NPN)	First-stage amplifier
Transistor Q2	BC516 (PNP Darlington)	High-gain LED driver
Indicator Diodes	6 × 1N4148	LED threshold setting
Indicator LEDs	4 LEDs	Visual signal level display

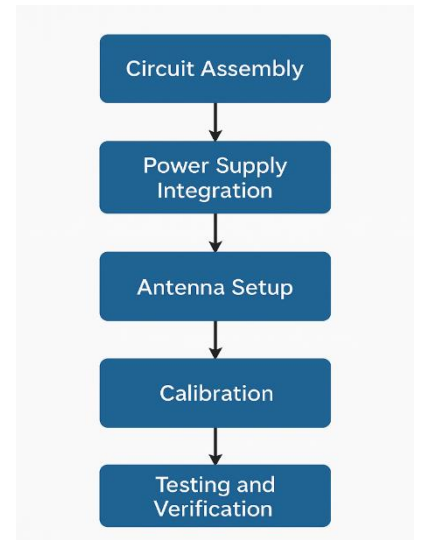
2 PROJECT DESIGN

2. Project Flowchart

Initial Power Test: After "Power Supply Integration", add a decision: *"Is the 12V supply stable?"*.

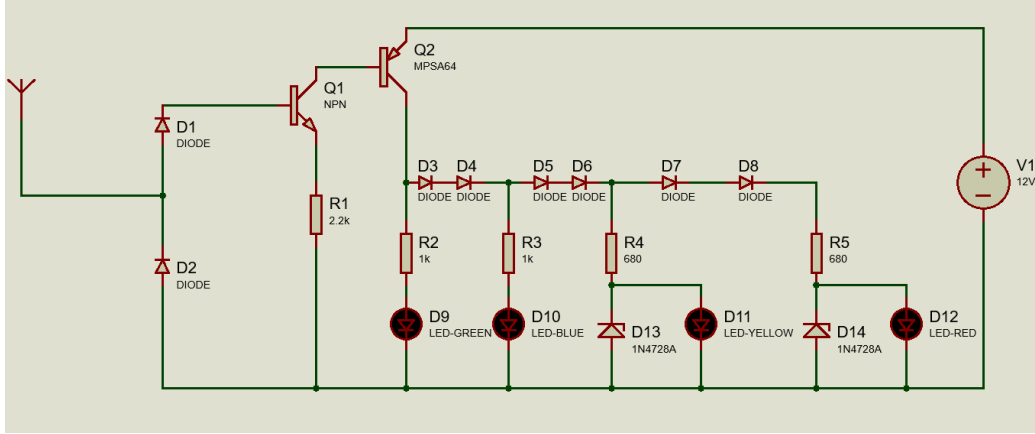
Sensitivity Tuning: After "Antenna Setup", add a decision: *"Is signal detected at the target frequency?"*. If not, loop back to "Antenna Optimization".

Calibration Verification: After "Calibration", add a check: *"Do LEDs activate in correct sequence (Low to Full)?"*.



Şekil 1: Flowchart of the project.

3. Schematic



Şekil 2: The schematic of the circuit.

4. Component Analysis

2.1 Circuit Analysis and Component Selection

The FSM operates from a 12 V DC supply and consists of three main functional blocks:

1. RF Detector
2. Two-Stage Amplifier
3. LED Indicator and Threshold Network

2.1.1. RF Detection and Rectification Stage

The 6-inch antenna acts as the primary RF sensor. The incoming RF energy is applied to a voltage doubler detector using two 1N60 germanium diodes. This configuration produces a proportional DC voltage that represents the strength of the received RF signal.

A 10 k Ω resistor couples this DC voltage into the base of the 2N4401 transistor, setting the input bias for the first amplification stage.

2.1.2. Two-Stage Transistor Amplifier

Stage 1 – 2N4401 (NPN)

Operates in a common-emitter configuration, providing initial voltage and current gain. Small changes in detector output cause corresponding changes in collector current.

Stage 2 – BC516 (PNP Darlington Pair)

The Darlington configuration provides extremely high gain. The collector current from the 2N4401 drives the base of the BC516, enabling the circuit to produce enough current to activate multiple LEDs under strong signals.

This high-gain architecture significantly increases the sensitivity of the FSM.

2.2. Indicator Stage and Calibration

The indicator stage consists of four LEDs, each corresponding to a signal strength threshold. A set of 1N4148 silicon diodes and series resistors define the current required to illuminate each LED.

Threshold Logic

Table 2: Threshold Logic Table.

Indicator	Resistor	Approx. Threshold	Meaning
LED1	1 k Ω	~5 mA	Weak signal
LED2	1 k Ω	~10 mA	Medium signal
LED3	680 Ω	~15 mA	Strong signal
LED4	680 Ω	~20 mA	Full-scale signal

The different resistor values ensure that LEDs illuminate progressively as the amplified signal increases.

5. Antenna Optimazation

In the design of an RF Field Strength Meter, the efficiency of the device is heavily dependent on the resonance of the antenna. For this project, a quarter-wavelength $\lambda/4$ monopole antenna was utilized. The relationship between the frequency of the signal and the required physical length of the antenna is governed by the following equations:

Fundamental Wave Equation

The wavelength λ in meters is determined by the speed of light c and the operating frequency f :

$$\lambda = \frac{c}{f}$$

Where $c \approx 300 \times 10^6$

Quarter-Wavelength Calculation

To achieve resonance for a monopole antenna, the length L is calculated as:

$$L = \frac{\lambda}{4} = \frac{c}{4f}$$

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PROJECT RESULTS AND IMPLEMENTAION

3.1 Implementation

The implementation phase involved translating the theoretical design into a practical, working prototype. The process was carried out in several steps:

Step 1: Circuit Assembly

The circuit was constructed on a solderless breadboard for initial testing. Key components included:

- RF Detector: Two 1N60 Germanium diodes configured as a voltage doubler to rectify the RF signal.
- Amplifier Stages: A 2N4401 NPN transistor for the first stage and a BC516 PNP Darlington pair for the second stage to achieve high current gain.
- Indicator Network: Four LEDs with series resistors and 1N4148 diodes to define threshold levels.

Step 2: Power Supply Integration

A regulated 12 V DC power supply was used to ensure stable operation.

Step 3: Antenna Setup

As documented in the experimental phase, the antenna was optimized for the UHF band:

- For $f = 443 \text{ Mhz}$:

$$L = \frac{300 \times 10^6}{4 \times 443 \times 10^6} = 0.169$$

This matches the optimized length of 16.9 cm used in the prototype to ensure maximum sensitivity.

An initial 6-inch wire antenna was connected to the detector stage. Later, antenna lengths were optimized for resonance at UHF (440 MHz) frequencies to improve sensitivity.

Step 4: Calibration

The LED indicator thresholds were calibrated by adjusting resistor values:

- LED1: Weak signal (~5 mA)
- LED2: Medium signal (~10 mA)
- LED3: Strong signal (~15 mA)
- LED4: Full-scale (~20 mA)

The indicator stage relies on a progressive current draw. By utilizing different resistor values $1k\ \Omega$ and $680\ \Omega$, the circuit defines current steps required to illuminate each subsequent LED:

- **Weak Signal:** Triggered at approximately 5 mA.
- **Full-Scale Signal:** Triggered at approximately 20 mA.

These thresholds allow the user to visually estimate the relative power of the RF field based on the number of active indicators.

Step 5: Testing and Verification

The prototype was tested using known RF sources:

- VHF Band (146 MHz): Verified antenna resonance and LED response.
- UHF Band (443 MHz): Confirmed sensitivity and LED progression. Measurements showed predictable LED activation as signal strength increased and antenna length matched the quarter-wavelength resonance.

Step 6: Final Assembly

After successful breadboard testing, the circuit was soldered onto a copper PCB for durability.

3.2. Experimental Results

The experimental results summarize the testing and verification phase of the RF Field Strength Meter.

Testing Sources:

UHF Band (440 MHz): Confirmed sensitivity and LED progression.

Observations:

- LED indicators activated predictably as signal strength increased.
- Antenna length adjustments improved sensitivity.
- Measurements showed consistent LED activation patterns for both UHF signals.

3.3. Project Conclusion

A fully functional RF Field Strength Meter was designed, constructed, and tested. The use of low-barrier 1N60 germanium diodes enabled excellent sensitivity to weak RF signals, while the two-stage transistor amplifier provided sufficient gain for LED level indication.

Key findings:

- Signal strength readings increased predictably as distance decreased.
- The meter successfully detected UHF signals.

3.4. Future Work

Possible enhancements include:

- Adding an analog or digital power meter to display results in dBm
- Integrating a potentiometer for adjusting the sensitivity.
- Designing housing enclosure.

REFERENCES

1. “Field Strength Meter Circuit,” Circuits-DIY.
2. Behzad Razavi, *RF Microelectronics*, 2nd ed., Pearson, 2012.
3. A. Kraus, *Antennas*, McGraw-Hill, 1988.
4. Horowitz & Hill, *The Art of Electronics*, 3rd ed., Cambridge University Press, 2015.